

$$2. \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{aligned} P_{\sigma_z} &= |\sigma_z\rangle\langle\sigma_z| \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

$$\begin{aligned} P_{\sigma_1} &= |0\rangle\langle 0| + |1\rangle\langle 1| \\ &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

3. eigenvector of σ_x

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{aligned} P_0(|+\rangle) &= \langle + | P_0 | + \rangle \\ &= \langle 0 | + \rangle^2 \end{aligned}$$

$$- P_r(|+\rangle) = \langle 0 | P_r | + \rangle = \langle + | 0 \rangle^2 = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{2} = 50\%$$

$$P_r(|-\rangle) = \langle 0 | P_r | - \rangle = \langle - | 0 \rangle^2 = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{2} = 50\%$$

5. We know that $U^\dagger U = I$
and $\hat{V}_i \hat{V}_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$

So when we consider a row of U

$$U: r_1, r_2, \dots, r_n$$

To prove orthonormality we need to show 2 condition

1. Normality: $\|r_i\| = 1$ for all i

2. Orthogonality: $r_i r_j = 0$ for all $i \neq j$

Proof Steps.

1. Take any row r_i to the inner product $r_i \cdot r_i = r_i^\dagger r_i = 1$

2. take any two different row r_i and $r_j \rightarrow r_i^\dagger r_j = 0$

$$6. \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{vmatrix} 0-\lambda_i & 1 \\ 1 & 0-\lambda_i \end{vmatrix} = 0$$

$$\lambda_i^2 - 1 = 0 \rightarrow \lambda_i = \pm 1$$

$$A|v\rangle = \lambda|v\rangle$$

$$A|v\rangle = \lambda I|v\rangle$$

$$|A - \lambda I| = 0$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \begin{vmatrix} 0-\lambda_i & -i \\ i & 0-\lambda_i \end{vmatrix} = 0$$

$$= \lambda_i^2 - (-i \cdot i) = 0$$

$$\lambda_i^2 - 1 = 0 \rightarrow \lambda_i = \pm 1$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{vmatrix} 1-\lambda_i & 0 \\ 0 & -1-\lambda_i \end{vmatrix} = 0$$

$$-(1-\lambda_i^2) = 0$$

$$\lambda_i^2 = 1 \rightarrow \lambda_i = \pm 1$$

8. We can take A as $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (σ_x)

We know pauli matrix are unitary.

We can prove it now

$$|\vec{v}_i\rangle = \frac{1}{\sqrt{2}}|\alpha\rangle - \frac{1}{\sqrt{2}}|\beta\rangle$$

$$|\vec{v}_i\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$A|\vec{v}_i\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

We can prove orthogonality by Adding square of coefficient of transformed vector

$$\left(-\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = 1$$

$$\left(\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}}\right) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) = 0$$